

MONARCH PASS, CO, USA

WMO: 722203

Lat: 38.497N

Lon: 106.319W

Elev: 3667

StdP: 64.36

Time Zone: -7.00 (NAM)

Period: 04-19

WBAN: 03041

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-22.0	-19.8	-27.4	0.5	-13.3	-25.1	0.6	-12.7	27.5	-10.2	25.3	-11.1	12.3	250	N/A

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	9.4	18.0	6.5	17.4	6.1	16.3	5.9	9.2	13.8	8.5	13.2	7.8	12.6	7.2	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7.5	10.2	9.9	7.0	9.9	9.6	6.1	9.2	9.1	37.9	13.8	35.9	13.2	33.6	12.2	11.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 23.4	21.3	19.3	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
			DB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
			WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	0.0	-9.4	-9.8	-5.8	-2.6	2.3	9.2	11.7	10.2	7.1	0.5	-4.4	-9.6
	DBStd	8.80	4.75	4.35	4.79	4.31	4.48	3.56	2.15	2.29	3.36	4.84	4.86	5.17
	HDD10.0	3779	602	554	489	377	241	54	9	26	93	296	432	607
	HDD18.3	6684	860	788	747	627	498	273	204	251	336	553	682	866
	CDD10.0	135	0	0	0	0	1	31	63	33	7	1	0	0
	CDD18.3	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDH23.3	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDH26.7	0	0	0	0	0	0	0	0	0	0	0	0	0
(14) Wind	WSAvg	9.9	11.4	12.9	11.5	11.1	9.0	8.4	6.3	6.0	8.0	10.4	11.6	12.4
(15) Precipitation	PrecAvg	297	16	16	22	29	26	18	40	48	31	22	15	14
	PrecMax	372	40	43	46	61	76	41	70	77	59	59	32	46
	PrecMin	166	2	5	5	7	2	2	14	14	5	6	1	5
	PrecStd	46	9	10	13	17	19	10	18	19	13	13	8	9
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	2.3	1.1	7.0	9.0	14.9	20.1	21.0	19.1	17.5	12.8	7.9	2.2
		MCWB	-3.9	-3.2	-0.9	0.1	3.1	6.6	7.0	6.9	6.1	5.5	0.9	-3.3
	2%	DB	-0.1	-0.9	3.9	7.3	12.5	17.8	18.8	17.4	15.9	10.9	5.0	0.1
		MCWB	-4.4	-4.5	-2.5	-0.8	2.3	5.5	7.0	6.3	5.5	4.1	-1.0	-4.1
	5%	DB	-2.0	-2.3	2.2	5.2	11.0	17.2	17.6	16.2	13.8	7.9	2.7	-1.4
		MCWB	-5.5	-5.4	-3.3	-1.5	1.8	5.0	6.7	6.2	4.6	1.9	-2.4	-4.7
	10%	DB	-2.7	-3.7	0.2	2.9	8.7	15.8	16.3	14.9	12.4	7.0	1.3	-2.6
		MCWB	-6.0	-6.8	-4.1	-2.5	1.1	4.6	6.8	6.3	4.2	0.7	-2.8	-5.9
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	-2.1	-2.1	-0.6	1.3	4.1	8.6	10.4	9.8	8.1	8.5	2.6	-1.4
		MCDB	-0.6	-0.5	5.3	6.5	11.6	14.5	14.9	14.6	13.2	10.4	6.6	0.7
	2%	WB	-3.4	-3.4	-1.7	0.2	3.2	7.3	9.5	8.8	6.9	5.0	-0.3	-3.2
		MCDB	-1.3	-1.4	2.8	4.5	10.2	14.8	14.1	13.2	12.2	8.4	3.1	-1.3
	5%	WB	-4.8	-5.1	-2.6	-0.7	2.4	6.4	8.9	8.1	6.3	2.9	-1.4	-4.4
		MCDB	-2.1	-2.8	0.9	3.3	8.5	13.7	13.7	12.4	11.4	7.0	1.8	-2.1
	10%	WB	-5.8	-6.1	-3.6	-1.5	1.6	5.6	8.2	7.3	5.5	1.6	-2.3	-5.5
		MCDB	-3.7	-4.3	-0.3	2.3	7.1	12.6	13.1	11.7	10.1	4.8	0.7	-3.5
(35) Mean Daily Temperature Range	MDBR		5.9	6.1	7.1	7.6	8.2	10.2	9.4	9.1	8.8	7.6	6.2	5.8
	5% DB	MCDBR	6.7	7.2	8.0	8.7	10.5	10.9	10.3	10.1	10.1	8.8	7.1	6.5
		MCWBR	4.7	4.9	4.5	4.8	4.8	4.9	4.1	4.2	4.3	4.5	3.9	4.1
	5% WB	MCDBR	6.1	6.6	7.4	7.8	9.3	10.4	9.1	8.7	8.5	7.9	6.6	5.7
		MCWBR	4.8	4.8	4.6	4.6	4.5	4.8	4.5	4.3	3.9	4.1	4.2	4.3
(40) Clear-Sky Solar Irradiance	taub		0.188	0.193	0.212	0.229	0.241	0.244	0.276	0.266	0.236	0.217	0.198	0.189
	taud		2.606	2.598	2.630	2.647	2.682	2.694	2.617	2.632	2.766	2.773	2.728	2.630
	Ebn at Noon		1051	1077	1072	1058	1040	1031	995	1003	1031	1035	1032	1028
	Edh at Noon		66	78	85	90	89	88	94	90	74	65	59	60
(44) All-Sky Solar Radiation	RadAvg		2.75	3.67	5.06	6.02	6.56	7.34	6.38	5.80	5.30	4.11	3.02	2.41
	RadStd		0.15	0.21	0.38	0.35	0.49	0.45	0.36	0.27	0.32	0.28	0.18	0.15

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (29 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+62	+16

Nomenclature: See separate page